

Particulate Matter Monitoring & Health Effects

Issue 2026-08-23 | Entry window 23 Aug 2026 | 9 in scope, 22 screened out

9

RECORDS IN SCOPE
(22 SCREENED OUT)

4

OF 10 CLUSTERS
OCCUPIED

2

ABSTRACT-FREE
DEPOSITS (22%)

0

RISK ESTIMATES
ON A PM CONTRAST

3

INSTRUMENT- OR
METRIC-LEVEL PAPERS

Signal of the day

A Sunday issue with no forest plot and an unusually direct message for the sensing side: the quantity being regulated and the quantity that drives inflammation are coming apart, and today's instrument paper is measuring the wrong one.

Zhong et al. (*Environ Int*) ran three years of Fuzhou observations against an animal-derived surface-area/inflammation dose-response and found that **6.6% of hours that satisfied the $PM_{2.5}$ standard ($<35 \mu g/m^3$) carried lung-deposited surface area $7.5\times$ above what the local LDSA/ $PM_{2.5}$ ratio predicts**, so that **82% of the predicted inflammatory burden was invisible to mass-based assessment**. In those hours median $PM_{2.5}$ was $3.0 \mu g/m^3$ — an order of magnitude under the standard — while the dose-response projects 27% lung neutrophil infiltration. Their ML attribution puts the mechanism on summer-autumn afternoon photochemistry producing 50–100 nm particles, i.e. a regime that *grows* as primary emissions fall and radiation load rises. Every optical low-cost sensor in the field, including the four characterised in today's other instrument paper (Werderman et al., *Atmos Pollut Res*), is a light-scattering mass proxy with essentially no response below ~ 300 nm. If Zhong et al. are right about the regime, the distributed-sensing literature is building dense networks for a metric that is becoming progressively less informative about the endpoint it is meant to proxy, and the missing instrument — a cheap, deployable surface-area or number-below-100 nm sensor — is not on anyone's roadmap in this corpus.

What changed today

- ▶ **The first four-way laboratory characterisation of the current low-cost sensor generation — and it arrived without an abstract.** Werderman et al. evaluate **Sensirion SEN66, Piersa IPS-7100, PurpleAir PIXEL and Alphasense OPC-N3** across aerosol types and concentration ranges. Three of the four are units this watch has not seen characterised before (the PIXEL is PurpleAir's move away from Plantower; the IPS-7100 is one of the few consumer-priced units advertising particle-number output). The publisher deposited title, authorship and DOI only, so it ships as *Metadata only* and is the highest-priority abstract retry on the queue. Last author Sousan ran the 2021 OPC-N3/SPS30/AirBeam2/PMS-A003 evaluation that established the environmental-vs-occupational calibration split, which is the lineage but not a finding.
- ▶ **A biomonitoring result strong enough to be a sensor.** Giannetti et al. sampled *Crematogaster scutellaris* at 30 traffic sites plus 4 controls in Parma and found magnetic susceptibility of particles on the ant exoskeleton tracking traffic intensity at **Spearman $\rho = 0.825$, $p < 0.001$** , with each additional **1,000 vehicles/day associated with a 1.41-fold increase** and roughly **700-fold across the sampled gradient**. Co-located soil showed nothing ($\rho = 0.167$, $p = 0.346$), which is the control that makes the result interesting: the ant is integrating the airborne fraction, not the deposited one. SEM-EDS puts Fe-oxide particles doped with Zn, Cu and Ti on the thorax, morphologically consistent with brake and tyre wear. No calibration to $\mu g/m^3$ and no CI on the slope — but as a zero-power, zero-maintenance spatial sampler it is a genuinely different point in the network-design space.
- ▶ **Mediation of the air-pollution/cognition link is still not identified.** Sakhvidi et al. find only **16 studies and 72 exposure-mediator-outcome associations** across six pollutants for the whole literature, with cardio- and cerebrovascular disease the most consistent candidate mediator and mental health, sleep and lung function unresolved. Reporting quality was judged good; risk of bias was not, particularly for outcome ascertainment and confounding. The honest reading is that the mediation question is under-powered rather than answered.
- ▶ **UK Biobank puts PM next to plasma neurodegeneration markers and finds the interaction mostly absent.** In 19,645 participants aged ≥ 50 with complete proteomic, metabolomic and environmental data, **GFAP was associated with dementia at HR 2.65 (95% CI 2.17–3.25)** and NfL with transitions to Parkinson's disease, dementia and death. **$PM \times NfL$ interacted with mortality only at the nominal level, and there was little evidence of PM interaction with either marker for dementia**; $NO_2/NO_x \times GFAP$ was *negative*. This is largely a null for the PM term and should be cited that way, not as support for a PM-neurodegeneration pathway.
- ▶ **No forest plot today.** The only CI-bearing estimate in the corpus has a plasma protein on its left-hand side, not a particulate contrast, and

Zhong's and Giannetti's numbers carry no intervals. f5 is omitted — the same call as 20 August.

Corpus analytics

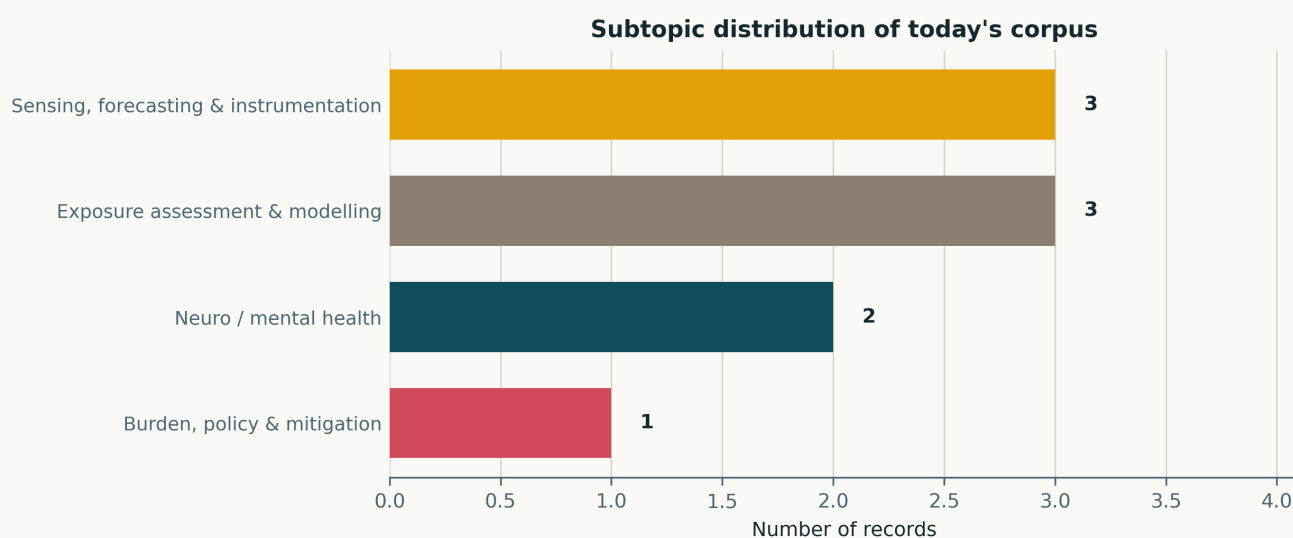


Fig. Subtopic assignment of the 9 in-scope records; single-label by dominant contribution. *Sensing, forecasting & instrumentation* and *Exposure assessment & modelling* take 3 each — **six of nine records sit on the measurement side of the field**, the most method-weighted day this watch has produced. **Only four of ten canonical clusters are occupied.** The six empty ones are mechanistic toxicology, cardiovascular/metabolic, respiratory/allergic, reproductive/developmental, occupational/indoor and other clinical endpoints — that is, the entire clinical half of the taxonomy is absent from a nine-record day.

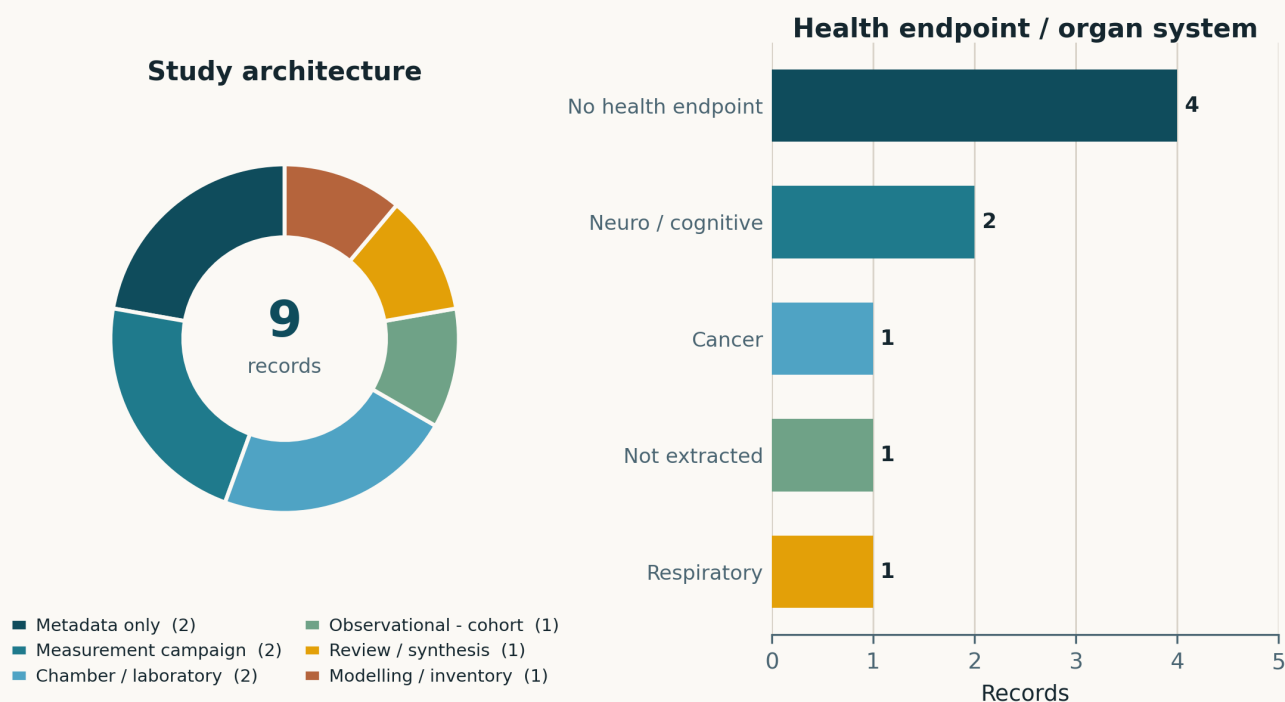


Fig. Left — study architecture, after grouping. The distribution is unusually flat: no group holds more than two of nine records — measurement campaign 2 (the Parma ant survey and the Fuzhou LDSA record), chamber/laboratory 2 (the CIMS calibration and the biomass-burning chamber-aircraft work), *Metadata only* 2, and one each of observational cohort, review/synthesis and modelling/inventory. The abstract-free share is 22%, less than half of Saturday's 46%. Right — health endpoint by organ system. Only four records carry a health endpoint at all (two neuro/cognitive, one respiratory, one cancer); the *No health endpoint* and *Not extracted* bars together are five of nine.

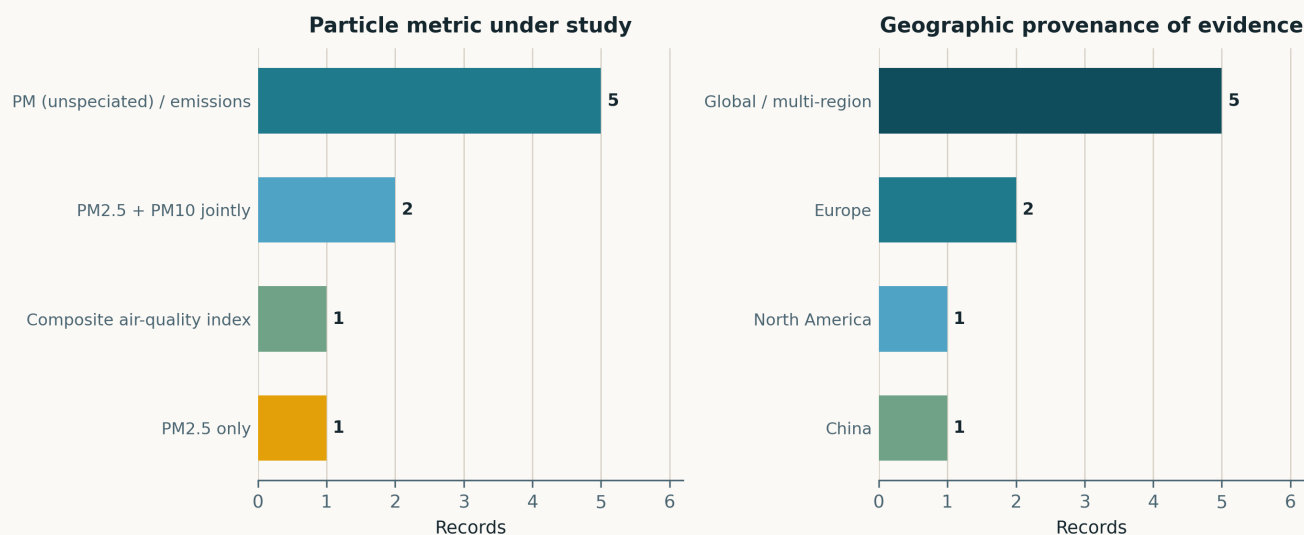


Fig. Left — particle metric. Note that only one record is PM_{2.5} mass in the conventional sense; the rest use lung-deposited surface area, magnetic susceptibility of deposited particles, aerosol oxidation state, OVOC precursor concentration, a composite microclimate index, or GBD risk-factor categories. **The day's metric diversity is itself the finding** and it is the direct visual counterpart to the LDSA/mass decoupling in the lead item. Right — geographic provenance. The *global / multi-region* bar of 4 is not four global studies: it is two genuinely global records (the GBD burden analysis and the cognitive-aging review), one two-country instrument campaign, and one metadata-only record whose study city the publisher did not deposit. Europe 2 (Parma, UK), North America 2, China 1.

Life-course windows addressed today (records may span several stages)

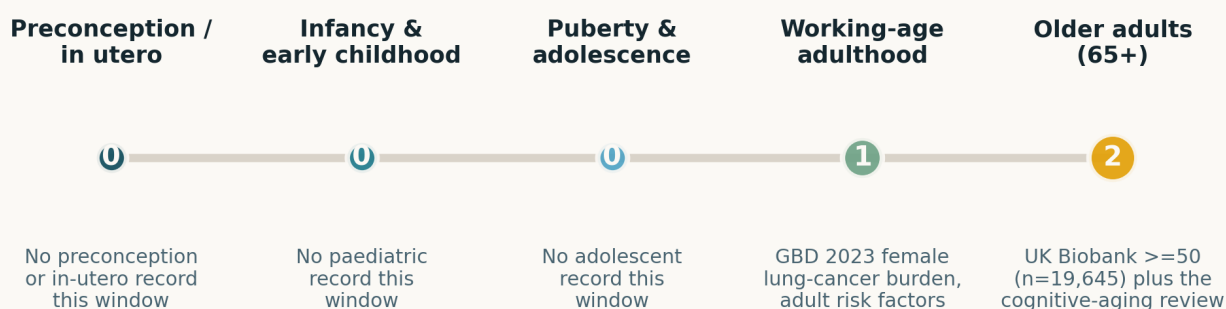


Fig. Life-course windows addressed, counted only where a record's study population is explicitly anchored at that stage. Two of five windows occupied and both at the old end — the UK Biobank ≥50 cohort and the cognitive-aging review at 65+, the GBD female lung-cancer burden across working-age adulthood. **No record this window studies a population under 18**, and the perinatal window, occupied yesterday by the aerosol-acidity preterm-birth stub, is empty again.

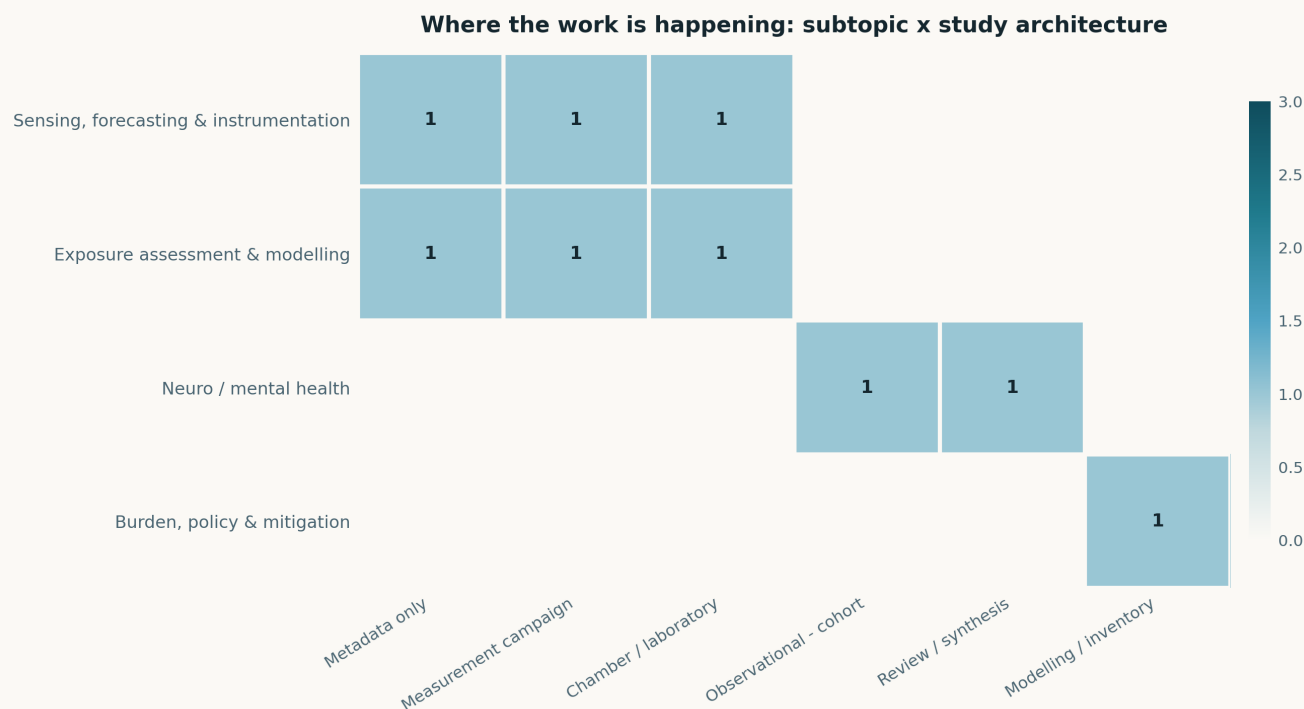


Fig. Subtopic × study-architecture cross-tabulation. With nine records across seven architectures the matrix is close to a permutation — no cell holds more than one record except *Metadata only*, which spans sensing and exposure modelling. There is no methodological concentration to report on a day this small, and reading a pattern into it would be over-interpretation.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase.

Sensing, forecasting & instrumentation

3 records

Werderman et al. 2026 Atmos Pollut Res

Laboratory evaluation of Sensirion SEN66, Piers IPS-7100, PurpleAir PIXEL and OPC-N3 low-cost aerosol sensors across aerosol types and concentration ranges

Metadata only. Crossref by-ISSN sweep; title, authorship (Werderman, Richards, Berkuta, Sousan) and DOI verified, no abstract deposited and no numbers extractable. It is nevertheless the single most on-topic record of the window: a chamber characterisation of *four* current-generation units across aerosol type and concentration range, three of which — SEN66, IPS-7100, PurpleAir PIXEL — have no prior evaluation in this corpus, against the OPC-N3 as the established reference point. The same last author's 2021 four-sensor evaluation established that environmental and occupational settings need separate calibration factors, so the plausible contribution here is the aerosol-type dependence of that factor for the newer optics. **Highest-priority full-text retrieval on the next run;** nothing is asserted about its findings until then.

doi: 10.1016/j.apr.2026.103185

Giannetti et al. 2026 Environ Pollut

Ants as bioindicators of urban traffic pollution

Thirty traffic-graded sites plus four controls in Parma; magnetic susceptibility and SEM-EDS on *Crematogaster scutellaris* exoskeleton and on co-located soil at 5 cm. **Ant magnetic susceptibility tracked traffic intensity at $\rho = 0.825$ ($p < 0.001$); soil did not ($\rho = 0.167$, $p = 0.346$).** On a log scale the relation is linear ($R = 0.66$, $p < 0.001$) with **1.41-fold per additional 1,000 vehicles/day**, ~700-fold across the gradient. SEM-EDS found Fe-oxide particles doped with Zn, Cu and Ti concentrated on the thorax, morphologically consistent with traffic-derived ultrafine PM. The weaknesses are that no slope CI is given, magnetic susceptibility is a proxy for the ferrous fraction only (it says nothing about organic or secondary mass), and nest-density confounding with traffic corridors is not addressed. For distributed sensing the interest is orthogonal: a passive, unpowered integrator with month-scale averaging at effectively zero unit cost, usable as a siting prior for where to put real instruments rather than as a replacement for them.

doi: 10.1016/j.envpol.2026.129010

Roska et al. 2026 Environ Sci Technol

Beyond targeted calibrations: comprehensive quantification of oxygenated organic compounds in urban atmospheres

A collision-induced-dissociation voltage-scanning method to derive sensitivities for an ammonium-adduct CIMS, so OVOCs without an authentic standard can still be quantified. Validated in the laboratory, then applied to the **2023 AEROMMA airborne campaign and the 2024 SAPHIR-CHANEL chamber campaign**: coverage rose to **98% of total signal in chamber and 78% in the field**, and adding the newly quantified species raised estimated **OH reactivity by ~5% and SOA formation potential by ~62%**. Coverage degrades with molecular size and oxidation state, and chamber-derived sensitivities are being transferred to ambient air on the strength of trend agreement rather than field standards — the paper says so. The 62% figure is the one that matters downstream: SOA yields used in chemical-transport models that feed exposure surfaces are calibrated against measurements that were missing most of this signal.

doi: 10.1021/acs.est.6c04149

Exposure assessment & modelling

3 records

Zhong et al. 2026 Environ Int*Photochemical ultrafine particles pose potential inflammatory risk despite low PM_{2.5} mass*

Three years of hourly Fuzhou observations; lung-deposited surface area compared against what the local LDSA/PM_{2.5} ratio predicts, with predicted inflammation from an animal-derived surface-area dose–response. **6.6% of PM_{2.5}-attainment hours (<35 µg/m³) carried 7.5× the expected surface-area exposure, and 82% of the predicted inflammatory risk in those hours was undetected by mass.** Median PM_{2.5} during them was 3.0 µg/m³, with the dose–response projecting 27% lung neutrophil infiltration. Machine learning attributes the events to photochemical production of 50–100 nm particles on warm, high-radiation summer–autumn afternoons (O₃ and secondary organic carbon dominate importance). **The limitation that threatens the inference is the dose–response itself:** an animal-derived surface-area relation extrapolated to a human population gives a predicted, not an observed, inflammatory response, and no human endpoint is measured here. Even so, the decoupling of LDSA from mass is measured, not modelled, and it is a direct challenge to the assumption that a dense network of scattering-based mass sensors indexes health-relevant exposure.

doi: 10.1016/j.envint.2026.110469

Zhang et al. 2026b Environ Sci Technol*Photoinduced intraparticle chemistry drives the rapid atmospheric oxidation of biomass burning organic aerosol*

Chamber isolation of biomass-burning aerosol aging pathways — UV-A irradiation, OH-driven SOA formation, heterogeneous OH oxidation, ozonolysis and evaporation — benchmarked against aircraft measurements of wildfire and prescribed-burn plumes. **UV-A irradiation drove intraparticle aging an order of magnitude faster than gas-phase oxidation**, at an estimated quantum yield of ~0.4% (four in a thousand absorbed photons generate particle-phase radicals). The authors estimate that **over half of oxygenated OA in biomass-burning-influenced regions may originate from photo-oxidised primary OA within the first day.** This inverts the standing assumption that gas-phase oxidation dominates BBOA aging, and it matters for exposure work because oxidation state governs both the oxidative potential of the inhaled particle and the optical properties that low-cost scattering sensors respond to — a smoke plume’s sensor bias is not constant along its trajectory.

doi: 10.1021/acs.est.6c02532

Mohammed 2026 Build Environ*Spatiotemporal dynamics of atmospheric vulnerability in hot arid cities: introducing the Urban Microclimate Air Stress Index (UMASI)*

Metadata only. Crossref sweep, no abstract deposited, no study city named in the metadata. A composite index combining microclimate and air-quality stress is exactly the kind of construct that needs its weighting scheme inspected before it is used, so nothing is characterised here. Flagged for retry; low priority relative to Werderman et al.

doi: 10.1016/j.buildenv.2026.115166

Neuro / mental health

2 records

Beydoun et al. 2026 *Ecotoxical Environ Saf**The interplay of air pollution with plasma neurodegenerative markers and metabolome for dementia, Parkinson's disease and all-cause mortality*

UK Biobank, **19,645 participants aged ≥ 50** with complete proteomic, metabolomic and environmental data; Cox models plus multistate Weibull models for transitions from healthy to dementia, Parkinson's disease and death. Higher NfL predicted all three transitions; **GFAP was specific to dementia, HR 2.65 (95% CI 2.17–3.25)**. On the exposure side the result is largely negative: **PM $\times$ NfL interacted with mortality only at the nominal level, NO₂/NO_x $\times$ GFAP interacted negatively, and there was little evidence of PM interaction with either marker for dementia**. Metabolomic principal components (lipid, amino-acid, energy) did modify the biomarker–outcome relations and interacted with PM for PD and dementia; a branched-chain-amino-acid component attenuated the GFAP–dementia association. The interaction tests are nominal and unadjusted for multiplicity across a large grid, so the positive ones should be treated as hypothesis-generating. Cite this for the biomarker result and for the *absence* of a PM interaction, not for a PM effect.

doi: 10.1016/j.ecoenv.2026.120657

Sakhvidi et al. 2026 *Ageing Res Rev**Ambient air pollution exposure and cognitive aging: mediators and possible pathways: a systematic review on the epidemiological studies*

16 studies, 72 unique exposure–mediator–outcome associations, six pollutants (PM_{2.5}, PM₁₀, NO₂, NO_x, black carbon, PM₁ components), with PM_{2.5} in 65% of analyses. Outcomes span memory, processing speed and diagnosed dementia; candidate mediators span depression/anxiety/stress, FEV₁/FVC/PEF, stroke and hypertension, CRP, type 2 diabetes, sleep and neuroanatomical change. **Cardio- and cerebrovascular disease emerged as the most consistent mediator**; mental health, sleep and lung function showed potential but unconfirmed mediation. Reporting quality was judged generally good, but most studies carried some risk of bias in outcome assessment and confounding adjustment. No pooled estimate, and 16 studies across six pollutants and this many candidate mediators is a thin base — the useful output is the map of which pathways have been tested at all, and the observation that almost none have been tested with a formal causal-mediation design.

doi: 10.1016/j.arr.2026.103330

Burden, policy & mitigation

1 record

Wang et al. 2026 *Front Glob Womens Health**Trends, inequalities, and 2030 projections of women's tracheal, bronchial, and lung cancer attributable to air pollution, radon, and tobacco exposure*

GBD 2023, female tracheal/bronchial/lung cancer 1990–2023, five inhalation-related risk factors, with SDI-gradient smoothing, concentration indices for inequality, decomposition by demographic vs epidemiological change, and projection to 2030. Smoking remains the leading contributor, **ambient particulate matter second, secondhand smoke third**. Over 1990–2023 **household solid-fuel-attributable mortality fell most sharply while particulate-matter-attributable mortality rose slightly**, with China the standout for the PM-attributable share. Along the SDI gradient, PM shows a **rising burden with diminishing inequality** — the opposite pattern to radon and tobacco — and decomposition attributes the PM increase to population growth and aging rather than to rising risk. Projected to keep rising to 2030. Standard GBD caveats bind hard here: the exposure is a modelled global surface, the attributable fraction rests on assumed counterfactual and RR functions, and the 2030 projection is an extrapolation of fitted trend, not a scenario.

doi: 10.3389/fghw.2026.1662451

Corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
Beydoun et al.	The interplay of air pollution with plasma neurodegenerative markers and metabolome for dementia, Parkinson's disease and all-cause mortality risks and transitions: the UK Biobank study	<i>Ecotoxical Environ Saf</i>	Cohort	Ambient PM _{2.5} / PM ₁₀ and NO ₂ /NO _x , modelled residential	United Kingdom
Sakhvidi et al.	Ambient air pollution exposure and cognitive aging; mediators and possible pathways: a systematic review on the epidemiological studies	<i>Ageing Res Rev</i>	Systematic review	PM _{2.5} , PM ₁₀ , PM ₁ components, black carbon, NO ₂ , NO _x	Global
Sensing, forecasting & instrumentation					
Giannetti et al.	Ants as bioindicators of urban traffic pollution	<i>Environ Pollut</i>	Field study	Magnetic susceptibility of traffic-derived PM on ant exoskeleton	Parma, Italy
Roska et al.	Beyond targeted calibrations: comprehensive quantification of oxygenated organic compounds in urban atmospheres	<i>Environ Sci Technol</i>	Instrument development	OVOC precursors of SOA; NH ₄ ⁺ -adduct CIMS	United States + Germany
Werderman et al.	Laboratory evaluation of Sensirion SEN66, Piera IPS-7100, PurpleAir P1XEL and OPC-N3 low-cost aerosol sensors across aerosol types and concentration ranges	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	Chamber PM across aerosol types (not extractable)	Not stated
Exposure assessment & modelling					

First author	Focus	Journal	Design	Metric	Region
Mohammed	Spatiotemporal dynamics of atmospheric vulnerability in hot arid cities: introducing the Urban Microclimate Air Stress Index	<i>Build Environ</i>	Metadata only (no abstract)	Composite microclimate/air-stress index (not extractable)	Hot arid cities (not stated)
Zhang et al.	Photoinduced intraparticle chemistry drives the rapid atmospheric oxidation of biomass burning organic aerosol	<i>Environ Sci Technol</i>	Chamber + aircraft	Biomass-burning organic aerosol oxidation state	United States
Zhong et al.	Photochemical ultrafine particles pose potential inflammatory risk despite low PM _{2.5} mass	<i>Environ Int</i>	Observational + ML	Lung-deposited surface area vs PM _{2.5} mass	Fuzhou, China
Burden, policy & mitigation					
Wang et al.	Trends, inequalities, and 2030 projections of women's tracheal, bronchial, and lung cancer attributable to air pollution, radon, and tobacco exposure	<i>Front Glob Womens Health</i>	Burden estimation	Ambient particulate matter and household solid fuel (GBD risk factors)	Global

Provenance and method

Entry window 2026-08-23 → 2026-08-23, contiguous with the 22 August issue. **This issue was built late**, on 24 August, because the scheduled Sunday run did not fire; the harvest window is unchanged and no gap was left, but the record is noted rather than concealed. **Sources.** PubMed E-utilities on [EDAT] returned **1 record** (a ruminant-methane study, screened out) — the lowest single-day PubMed yield the watch has recorded, consistent with a Sunday deposit floor; the connector re-query on both the health and the instrumentation axis returned zero for the same window. Europe PMC on CREATION_DATE returned 20 records, of which **13 were archival PMC back-scans** (publication years 1974–2005, no title in the record) and 5 shipped. The Crossref by-ISSN sweep returned 12 records after front-matter filtering and shipped 4. **OpenAlex returned HTTP 429 for the 21st consecutive run.** arXiv returned nothing new under the 3-day recency screen. Consensus and Scholar Gateway were both queried for the Werderman et al. sensor evaluation and neither has indexed it yet, which is why it ships as a stub. **Screening.** 22 records were screened out and written to `state/rejected.jsonl` with an individual reason each; the 13 archival back-scans carry a shared structural reason. One borderline exclusion is logged as `borderline`: a national wastewater-based-epidemiology assessment, environmental and population-scale but with no airborne route. **Labels.** Two records carry no abstract and are marked *Metadata only*; their design, sample size, endpoint and study location are recorded as not extracted rather than inferred. **No forest plot** is included — see the closing bullet of “What changed today” for why. All DOIs were checked by `check_dois.py` before publication. Content derived from publisher metadata; abstracts remain the copyright of their respective publishers.